



Multi-View XAI Dashboard

Unified Diabetes Risk Prediction with Explainable AI & Medical Narratives



Model Performance Comparison

ANN Regression (Progression Score)

RMSE: 0.478

MAE: 0.352

R² Score: 0.892

Status: ✓ Excellent Fit

Logistic Regression (Risk Classification)

Accuracy: 0.915

Precision: 0.902

Recall: 0.927

F1-Score: 0.914

AUC-ROC: 0.968

Status: ✓ High Sensitivity



Comparative Analysis Panel

Metric

ANN

Logistic

Primary Task

Regression

Classification

Metric	ANN	Logistic
Output Type	Continuous Score	Binary Risk Level
Model Complexity	High (Non-linear)	Low (Linear)
Interpretability	Medium (XAI Required)	High (Intrinsic)
Training Time	Longer	Instant
Clinical Use	Risk Quantification	Risk Stratification
Best For	Detailed Progression	Binary Decisions



Explainable AI Methods Applied

SHAP

Game Theory Approach

Feature importance based on Shapley values. Global and local explanations.

Global + Local

LIME

Local Linear Approximation

Model-agnostic explanations for individual predictions via local surrogate.

Local Only

Captum

Attribution Methods

Neural network-specific attribution through gradient and perturbation analysis.

Network-Native



Top Feature Importance (SHAP)

Glucose Level

95%

BMI (Body Mass Index)

87%

Age

78%

Blood Pressure

72%

Insulin Levels

64%



LLM-Generated Medical Narrative

Patient Risk Summary:

This patient presents a moderate-to-high risk profile for diabetes progression. The dual-model analysis indicates a predicted diabetes progression score of 6.8/10 with high-risk classification (92% probability). This assessment is driven primarily by elevated glucose levels and elevated BMI, which collectively account for 82% of the model's decision.

Key Risk Drivers (SHAP Analysis):

- Glucose Level (95% importance):** The patient's fasting glucose is significantly elevated at 147 mg/dL, representing the single strongest predictor of diabetes risk. Each 10 mg/dL increase above normal baseline contributes ~0.35 points to the progression score.
- BMI (87% importance):** At 34.2 kg/m², the patient falls into the obese category. BMI shows strong non-linear relationships with diabetes risk, particularly in the 30+ range where risk accelerates.
- Age (78% importance):** At 52 years old, the patient enters a high-risk demographic window. Age-related metabolic changes compound glucose metabolism challenges.

LIME Local Explanation:

When examining this specific prediction, LIME reveals that for this patient's feature profile, a 5-point increase in glucose would push the risk probability from 92% to 98%, while a 2-point BMI reduction would lower it to 87%. This local sensitivity analysis suggests that glucose management is the most actionable intervention point.

Captum Attribution Insights:

Deep network analysis shows that glucose information propagates through multiple hidden layers with amplifying effect, suggesting the ANN has learned complex non-linear interactions between glucose and other metabolic markers (insulin, BMI). The model distinguishes this patient from low-risk profiles with high confidence.

Clinical Recommendations:

- Implement intensive glucose monitoring (HbA1c check every 3 months)
- Recommend structured weight loss program targeting 5-10% reduction
- Consider lifestyle intervention before pharmacological treatment
- Monitor blood pressure as secondary preventive measure
- Retest in 6 months to track progression trend

Model Confidence:

Both ANN ($R^2=0.892$) and Logistic Regression ($AUC=0.968$) demonstrate high predictive power on this patient cohort. The concordance between regression score (6.8/10) and classification probability (92%) provides confidence in this risk assessment. All three XAI methods converge on glucose and BMI as primary drivers, supporting explanation robustness.

Multi-View XAI Tool | Diabetes Risk Prediction System

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